

Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents

Xiaoxuan Lei

(co-authors TBD)

Draft v0.7 — IAB: Interpreting Agent Behavior, Workshop @ NeurIPS 2026

Abstract

More interaction does not make LLM agents better lateral thinkers. We ran 693 games of Turtle Soup—Chinese situation puzzles solved by asking yes/no questions—across three model scales and thirty rounds of questioning. Accuracy after each round is flat from the first round to the thirtieth, for every model. A larger round budget changes nothing for the two larger models and actively hurts the smallest. Flat curves say interaction is not helping; they cannot say why. So we read the transcripts instead. Embedding each question and tracking it over the dialogue gives two signals—how far the search moves each round, and in what direction—and those separate failure modes that identical scores hide. The 4B model *stalls*: its late-game questions move 0.003 in semantic space, circling one hypothesis it never commits to. The 397B model *explores without banking*: it strides 0.16 per round, covering real ground, and still never converts any of it into a better story. Both score zero. We also report what the geometry does not do: a mixed-effects analysis shows puzzle identity absorbs its predictive signal ($p = 0.28$), so we offer it as an instrument for reading behavior, not for forecasting success. Finally, a warning for anyone building benchmarks of this shape: a weak answering model quietly makes the task unsolvable, so what looks like agent failure can be environment failure.

1 Introduction

Here is a Turtle Soup puzzle. *A man’s body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story is that a hot-air balloon was losing altitude, the passengers threw out their luggage and still could not stay up, so they drew lots with matchsticks—and the man who drew the short one was thrown out. The solver recovers this by asking yes/no questions, one per round.

Give a language model thirty such rounds and it does not get better as the dialogue grows. On our 693-game grid, accuracy after each round is flat from round 1 to round 30 for every model we test, and a larger round budget helps no one—it actively hurts the smallest model. Outcome metrics can state this flatness. They cannot explain it. Two agents scoring 0.0 on the same puzzle may be failing in opposite ways: one never leaves its first hypothesis, the other leaves the plausible region entirely. Explaining the flatness needs instruments that read the process, and that is this paper’s contribution.

Why Turtle Soup. Three properties make it a good behavioral microscope. The questioning protocol is a naturally verbalized search trace—no probing or activation access needed, the behavior *is* text. Puzzles are deliberately under-determined, so the direction of each associative jump is diagnostic. And games are short, language-only, and cheap to run at scale.

The gap. Prior turtle-soup benchmarks score verification correctness, solve efficiency, or consistency with the single intended story [1, 2, 3]. None interprets the search behavior itself.

Central hypothesis. An agent’s per-round questions, embedded in a semantic space, form a trajectory whose geometry explains success and failure.

- **H1 (stalling).** An agent that cannot leave a hypothesis basin shows step size going to zero: successive questions are semantically near-identical. Lexical novelty heuristics catch only the verbatim extreme of this.
- **H2 (drifting).** An agent that explores in directions no human path takes shows normal step size but growing distance from a human association anchor. Lexical novelty misses this entirely, because synonym-shuffled re-asks look new.
- **H3 (predictivity).** Trajectory features predict final solve quality beyond outcome-adjacent covariates, and the two failure modes have distinct signatures.

Contributions. (1) The flatness result: across three model scales, interaction length buys no accuracy and degrades the smallest model. (2) A behavioral-interpretation framework that explains the flatness by separating stalling from unbanked exploration, with an honest account of what the geometry does and does not predict. (3) An open-source

harness, a 25-puzzle benchmark restricted to puzzles humans have played and solved, and a composite score that separates clue recall from judged causal logic. (4) A methodological finding: Oracle competence must be audited first, or agent scores measure the environment.

2 Related Work

TurtleBench [1] collects 1,532 real user guesses and evaluates models as yes/no judges—verification, not exploration. SPLAT [2] introduces 975 graded puzzles and the interactive player–judge protocol we adopt, scored by accuracy and rounds. TurtleSoup-Bench [3] frames 800 bilingual puzzles with multi-dimensional scoring, still rewarding faithfulness to the single intended story. All three score the destination; we score the trajectory.

MUTATE [4] shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow agent is a natural intervention to test against our metrics. Our instrument adapts two human-psychometric ideas: forward flow [7], which measures how far free-association chains travel and predicts creativity, and the Divergent Association Task [8], where semantic distance among generated words is itself the measure. The Small World of Words norms [9] would supply an empirical human anchor—§3 says why ours is not yet that. Judge-bias catalogues [5, 6] shape our scoring: the model judge is confined to the one subscore objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 and asks one yes/no question per round. An **Oracle** holds the 汤底 and answers 是/不是/与此无关 (yes/no/irrelevant). The Questioner may commit a final story at any round, or is forced to at the budget. The harness has pluggable providers—local, gateway, and token-billed sampling including RL-trained checkpoints—with full per-round logging and per-game seeds.

3.2 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice or that its intended leap is findable, so an agent failing it teaches us nothing. Our benchmark takes puzzles with a public record of human play: 25 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 35), solutions 13–279 (median 102), with 115 annotated key clues (median 5 per puzzle). Provenance is enforced in code—a verified family and a quarantined generated family, with tests that fail if they mix.

3.3 Outcome scoring: composite, not surface

String matching cannot tell a wrong answer from a right one worded differently. We watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Our composite score is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches per-puzzle annotated clues: objective and reproducible. The logic judge rates only whether cause → mechanism → outcome matches, is told explicitly to ignore wording, and is sampled $k=3$ and averaged, since single ratings are unstable on borderline answers. The clue half also blocks “far = creative” gaming: distance without validity scores zero.

3.4 Association-trajectory instrumentation

Each round, we extract the question’s keywords, embed them, and aggregate to a unit vector q_t . Two signals follow: **step size** $s_t = d(q_t, q_{t-1})$, the agent’s associative stride, and **anchor distance** $h_t = d(q_t, \mathcal{A})$. Hypothesized signatures: stalling is $\bar{s} \rightarrow 0$ with h flat; drifting is normal $\bar{s}$ with h_t rising.

On the anchor. Ours is built from the puzzle’s surface, solution, and clues, and we report that plainly rather than call it a human manifold. Two consequences we measured: 41–69% of its terms come only from the solution and clues, and clue recall is the outcome, so the two are partly definitionally linked; and rebuilding the anchor from the surface alone flips the sign of the drift slope in 4 of 11 held-out traces. We report both anchors and treat a SWOW-based version [9] as the correct future form.

4 Experiments

E1 — Round curve. Play to 30 rounds; after each round force a checkpoint answer and score it. Does interaction help, and where does behavior degrade?

E2 — Round budget. Hard caps $\{5, 10, 15, 20, 25, 30\}$ with a forced final answer. How does behavior change under pressure?

E3 — Association trajectory. For every E1 game compute $\{s_t\}$ and $\{h_t\}$ and test H1–H3. The keyword extractor is held fixed across models, so no model is scored through its own vocabulary. We report under two encoders, and keep

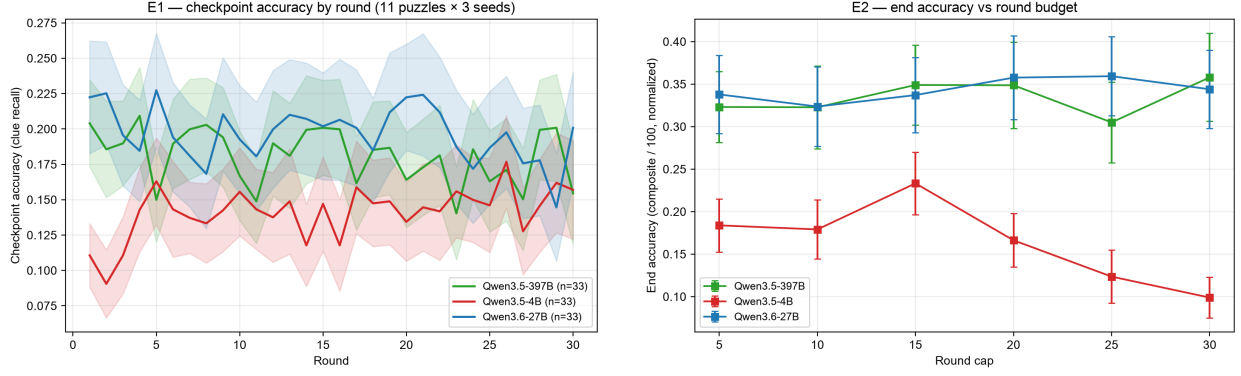


Figure 1: E1 (left): accuracy after each round, flat across 30 rounds for all three models (mean $\pm$ SEM, 33 games each). **E2 (right):** accuracy against round budget—the two larger models flat at 0.32–0.36, the 4B peaking at a budget of 15 and then degrading to 0.10.

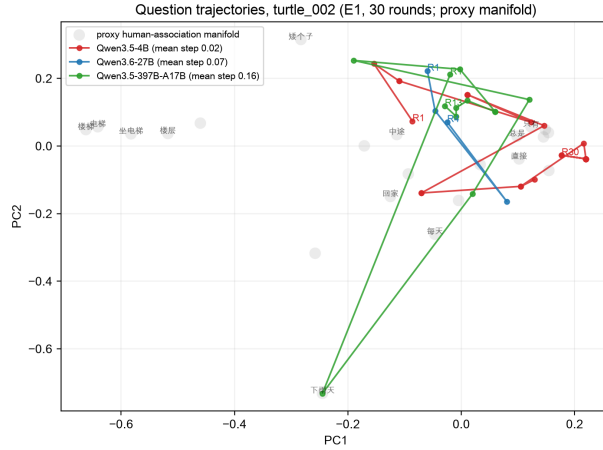


Figure 2: Two ways to score zero, on one puzzle (E1, seed 0; questions projected to 2-D; shaded region: the association anchor). The 4B (red) plays all 30 rounds inside a tight loop—late-game step size 0.003—and never commits. The 397B (green) strides at 0.16 for thirteen rounds before narrowing to 0.02 as it converges: explore-then-commit. The 27B (blue) lands on the right hypothesis at round 1 and commits by round 4. *This puzzle was hand-added rather than sourced from the community site and is no longer in the released set (§3); the figure illustrates the signatures and will be regenerated.*

the lexical question_novelty metric as the ablation baseline H2 predicts will fail: it catches verbatim repetition but not synonym-shuffled re-asks of the same hypothesis.

E4 (exploratory). Diverge-then-narrow prompting [4]—which trajectory feature does it move? And do signatures shift on puzzles admitting many distinct stories versus one?

Prerequisite — the Oracle audit. The Oracle is the sole information channel, so its competence caps what any Questioner can learn. In an early pilot ours was a 4B model that answered 3 of 10 probe questions correctly, with all seven errors collapsing to “irrelevant”—the one reply carrying no information. Raising the Oracle to 90% and then 100% left the Questioner’s score at 0.00 throughout: the environment had been a bottleneck, and removing it revealed a second one. Every experiment below fixes an Oracle above 90%, audited on held-out probe pairs before any agent claim.

5 Results

Scope. 693 games (99 E1, 594 E2): 11 puzzles $\times$ {4B, 27B, 397B} $\times$ 3 seeds, strong fixed Oracle. This grid predates the puzzle-set rebuild in §3 and has no overlap with it, so we report it as preliminary evidence of the phenomena and of what the instruments detect.

E1: interaction does not improve the working hypothesis. Accuracy is flat across all 30 rounds for every

model (Fig. 1, left): 27B ≈ 0.20 , 397B ≈ 0.18 , 4B ≈ 0.14 , with SEM bands overlapping between round 1 and round 30 throughout. What does separate the models is when they will commit. The 4B never volunteers a final answer in 33 games. The 27B commits readily (9/33, mean round 8.7). The 397B commits rarely and late (2/33, mean round 19.5). Calibration discriminates where accuracy does not.

E2: budget buys the small model degradation, not accuracy. The 27B and 397B sit at 0.32–0.36 from a cap of 5 to a cap of 30. The 4B peaks at cap 15 (0.23) and then declines monotonically to 0.10, with 29 of 198 games dying by token exhaustion. No model converts extra rounds into accuracy; the smallest converts them into noise.

E3: the geometry separates the two failures. Over 99 traces, movement away from the anchor co-occurs with failure ($r = -0.28$ pooled, -0.43 for the 27B), subject to the anchor caveat above. Stride is capacity-dependent: pooled it correlates with nothing, but per model it inverts with scale—4B $\rho = -0.46$, where large jumps are incoherence, and 397B $r = +0.21$, where they are exploration. Raw step size is not a health signal on its own; step size read against capability tier is. At the trace level the signatures are unmistakable (Fig. 2): the 4B’s late-game stride of 0.003, the 397B’s 0.16 for thirteen rounds narrowing to 0.02 as it converges, and a 27B run that lands on the right hypothesis at round 1 and commits by round 4.

H3 is not supported, and that is informative. Trajectory features add no significant predictive power *within* a puzzle (mixed-effects, puzzle random intercept, $n = 99$: drift $\chi^2 = 1.16$, $p = 0.28$; stride $\times$ tier $\chi^2 = 1.29$, $p = 0.73$), while the puzzle random-intercept variance is large (0.050 on a 0–1 outcome). A pooled correlation splits into a within-puzzle term, which is about the agent, and a between-puzzle term, which is about difficulty—and hard puzzles both induce drift and defeat agents, producing the pooled correlation for free. A metric can separate behaviors reliably while predicting outcomes poorly. We therefore claim the geometry as an interpretive instrument, for reading *how* an agent is failing, not for forecasting whether it will.

Interpretation. Flat outcome curves are the shadow of two different behaviors. Small models stall and degrade with budget. Larger models explore but do not bank exploration into an improving hypothesis. Neither diagnosis is visible in any outcome metric.

6 Pilot Observations

Four findings from the harness, each showing the framework detects something real—and three of them are traps for anyone building a benchmark of this shape.

Stalling is real, and lexical metrics only half-see it. A 4B model re-asks earlier questions *verbatim* in later rounds. A lexical novelty metric catches that—but only because it is verbatim. Earlier the same model re-asks one hypothesis in fresh words, which that metric scores as new and the geometry does not.

Surface scoring misleads; the composite fixes it. A frontier model reconstructed a hidden story’s full causal chain and scored 0.00 under clue-string matching. The composite separated its logic (30/30) from its terminology (28/70)—and that gap is itself interpretable: right story, wrong vocabulary.

Truncated thinking masquerades as behavior. Under a 512-token budget, a hybrid reasoning model’s logged “questions” were fragments of chain-of-thought, and every Oracle reply to them was “irrelevant.” From the transcript this is indistinguishable from an agent that cannot form a question. Disabling template thinking recovered an 8-round solve. Decoding configuration is part of the environment.

Linearly solvable puzzles are poor probes. A strong model solved a classic puzzle in five straight-line rounds—no reframing, no dead ends. Puzzles like this cannot discriminate between agents, which is what motivates grading the item set by how much of the solution the surface leaves under-determined.

7 Limitations and Conclusion

Limitations. The grid predates the puzzle-set rebuild and must be re-run on the released 25 items; we say so rather than narrow the claims to fit the data in hand. The anchor is not yet human-derived. The coordinate that would settle stalling versus unbanked exploration most directly—whether the next question depends on the last answer—is definable but unmeasured. In the pilot grid the largest model was Questioner, Oracle, and judge: the answering role is mechanical, the judging role is not. Puzzles carry dark themes (content-flagged) and the source site is public, so memorization probes are required.

Conclusion. Turtle Soup turns hypothesis-space search into observable text, and trajectory geometry turns that text into behavioral signatures. An endpoint score projects that search onto one number, destroying what you need to improve an interactive agent: one that stalls and one that explores without banking need opposite interventions. Reading the trajectory tells you which you have.

References

- [1] Q. Yu *et al.* TurtleBench: Evaluating Top Language Models via Real-World Yes/No Puzzles. arXiv:2410.05262.
- [2] Q. Chen, B. Zhang, G. Wang, Q. Wu. Weak-eval-Strong: Evaluating and Eliciting Lateral Thinking of LLMs with Situation Puzzles. arXiv:2410.06733.
- [3] M. Zhou *et al.* What to Ask Next? Probing the Imaginative Reasoning of LLMs with TurtleSoup Puzzles. arXiv:2508.10358.
- [4] J. Park, I. Baek, J. Park, H. Lee. Beyond One Path: Evaluating and Enhancing Divergent Thinking in Interactive LLM Agents. arXiv:2605.28465.
- [5] J. Ye *et al.* Justice or Prejudice? Quantifying Biases in LLM-as-a-Judge. ICLR 2025. arXiv:2410.02736.
- [6] L. Shi *et al.* Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. IJCNLP-AAACL 2025.
- [7] K. Gray *et al.* “Forward Flow”: A New Measure to Quantify Free Thought and Predict Creativity. *American Psychologist* 74(5), 2019.
- [8] J. A. Olson *et al.* Naming Unrelated Words Predicts Creativity. *PNAS* 118(25), 2021.
- [9] S. De Deyne *et al.* The “Small World of Words” English Word Association Norms. *Behavior Research Methods* 51(3), 2019.

A The validation set

Where the puzzles come from

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable at all. An agent scoring zero on such an item tells us nothing about the agent. We therefore take only puzzles with a public record of human play: items tagged 经典 (“classic”) on a Chinese Turtle Soup community site, each keeping its source URL.

Two rounds of human review followed. The first cut eleven items—one was a deduction puzzle presenting suspect testimonies rather than a lateral one, several had solutions long enough to need 32k-token budgets, and three had surfaces the site itself had damaged (a long surface truncated to a placeholder with the real text buried in the solution field; a scrape artefact and an author’s preamble prepended). Ten short puzzles replaced them and a further item was cut on a second pass. The site’s 经典 tag is exhausted, so the ten replacements carry other tags; the boundary is stated rather than assumed. What remains is 25 items: surfaces of 9–108 characters (median 35), solutions of 13–279 (median 102), and 115 annotated key clues (median 5 per puzzle).

Provenance is enforced in code rather than by convention—a verified family and a quarantined generated family, with tests that fail if the two ever mix. This matters because the grid in §5 ran on an earlier set in which only two of eleven puzzles were real: six were model-generated (one still carrying a development placeholder as its title, three near-duplicate variants of a single motif) and three were written by us and never tested on anyone.

One game, in full

Surface (汤面). 沙漠里躺着一具男尸，手里紧紧攥着半根火柴，周围散落着一些行李箱、衣物。(A man’s body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.)

Base (汤底), hidden from the Questioner. A hot-air balloon crossing the desert began losing altitude. The passengers threw out their luggage and were still too heavy, so they drew lots with matchsticks; the man who drew the short half was thrown out and fell to his death.

Questioner (sees only the surface)	Oracle
1 Did he die from falling from a height?	是
2 Before falling, was he airborne in some craft—a balloon or a plane?	是
3 Was it a hot-air balloon?	是
4 Were there passengers besides the victim?	是
5 Did they throw luggage and clothing out to lose weight?	是
6 Still too heavy after jettisoning everything, did they draw matchsticks to decide who would jump—and he drew the short half?	与此无关

Round 6 states the hidden story completely and is answered “irrelevant.” The Oracle is wrong: re-running that question returns 是 twice out of three samples. One example therefore carries three things at once—the protocol, a converging trajectory (five confirmations narrowing steadily), and a reminder that the environment is a model and makes mistakes at the moment they cost the most.

How difficulty is judged

The mixed-effects result in §5 says puzzle identity dominates outcome without saying *what* about a puzzle does the dominating. A difficulty measure that could supply the account needs two properties: it must describe the puzzle rather than an agent’s performance on it, and it must be independent of rounds used—otherwise grouping an accuracy-versus-rounds plot by it would make the plot confirm itself.

We measure *under-determination*: how far the surface alone leaves you from the solution. Given only the 汤面, a model produces $N=12$ complete stories, each required to explain every detail of the surface and to differ from the others in mechanism rather than in names or setting. It never sees the 汤底, which would anchor it on the true answer. Each story is scored by embedding similarity to the real solution, and the index is $1 - \max_i \text{sim}(\text{story}_i, \text{solution})$: large when no cold guess comes close.

Why similarity rather than clue matching. Our first version scored cold guesses by key-clue recall and rated the balloon puzzle 0.33—although its very first candidate was the correct story. The candidate said “drawing matchsticks to decide who jumps” where the annotated clue reads “drawing lots”; a two-character clue is too short for the bigram fallback, so the match failed. That version measured vocabulary, not difficulty.

Why one dimension rather than two. We first tried to separate *depth* (the solution sits far from the surface, taking many confirmations to reach) from *breadth* (the surface admits many stories, so progress comes by elimination). Mea-

sured separately both saturated—every puzzle scored 4 of 5 on a depth rubric, and distinct-candidate counts sat at their ceiling. They are not independent either: a surface loose enough to admit many mechanisms is, for that same reason, one whose single true mechanism takes more reframing to reach. Under-determination absorbs both.

Approaches that failed, recorded so they are not repeated: surface-to-solution embedding distance (uncorrelated with observed accuracy, and it rated the puzzle a strong model solved in five straight rounds the hardest in the set); adding annotated clues to the surface one at a time and watching the distance fall (it does not fall monotonically—several clues push it up); a 1–5 depth rubric (no calibrated scale, so everything lands mid-range); counting unexplained surface details (bounded at 1–3 by how few odd details a short surface can hold, though still useful for saying *what* makes a particular puzzle hard).

How accuracy is measured

String matching cannot tell a wrong answer from a right one worded differently: we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. The composite score separates what objective matching can verify from what it cannot.

Clue recall (70). Annotated key clues matched against the final answer, with a character-bigram fallback so that sentence-form clues do not collapse the match to exact substring. Clues are model-extracted under three constraints that make them scorable: each must be a causal element of the solution, must *not* already appear in the surface (or it is free points), and must be grounded in the solution’s own vocabulary (or it is invented). The earlier lexicon-based extractor matched a word list written for the first ten puzzles, so on any other puzzle it emitted mid-sentence fragments and words that fit any story at all—seventy points scoring noise.

Causal-logic identity (30). A judge rates only whether cause → mechanism → outcome matches, instructed explicitly to ignore wording, sampled $k=3$ and averaged because single ratings are unstable on borderline answers. The gap between the two subscores is itself readable: right story, wrong vocabulary.

Difficulty across the set

[To be filled once all 25 puzzles are annotated: the distribution of under-determination, and its correlation with surface length, clue count, and observed accuracy.]

A caveat we report either way. The two puzzles scoring lowest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure is reading familiarity as much as difficulty. A memorisation probe—asking for the solution with no surface given—separates the two and belongs in any use of this measure.